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Aaron Thomas

Profile

Final year Computer Science student building production backends. REST APIs in Python/FastAPI and Node.js, PostgreSQL and Firebase data models, and AWS deployments, from government CMS and LMS platforms to Android app backends.

Technical Skills

- **Languages:** Python, Go, Rust, TypeScript/JavaScript, Java, C
- **Backend:** Node.js, FastAPI, REST APIs, PostgreSQL, MongoDB, Firebase, Supabase
- **Frontend:** React, Next.js, React Native, Tailwind CSS
- **Cloud/DevOps:** AWS, GCP, Docker, CI/CD
- **AI/ML:** LangChain, LangGraph, LlamaIndex, Prompt Engineering
- **Soft Skills:** Communication, Teamwork, Leadership, Problem Solving

Experience

Backend Developer (Contract) — **Dubai, UAE (Remote)**
Lions Information Technology L.L.C — *04/2026 – 08/2026*
- Built backend services and REST APIs powering Android applications using Python and FastAPI.
- Designed PostgreSQL data models and deployed services on AWS for mobile clients.

Software Engineering Intern — **Thiruvananthapuram, KL**
Stride, K-DISC — *10/2025 – 10/2026*
- Developed enterprise CMS and LMS platforms for the Government of Kerala using Next.js, Node.js, and AWS.

Backend & Automation Engineer (Part-time) — **Kodakara, KL**
Sahrdaya College of Engineering & Technology (Autonomous) — *08/2025 – 02/2026*
- Architected backend systems for the official college website and CMS using Firebase, Node.js, and REST APIs.
- Built workflow automation and system orchestration to streamline CMS operations and reduce repetitive manual work.

Web Development Intern — **New Zealand (Remote)**
Pexcel Solutions — *12/2024 – 06/2025*
- Developed responsive frontend interfaces using React.js and Tailwind CSS for international client projects.

Projects

Aliens Eye — github.com/arxhr007/Aliens_eye
AI-Powered Cross-Platform Profile Analysis
- Built an async platform analyzing public profiles across 840+ sites with ML/heuristic scoring and cross-site profile correlation.
- Achieved **4700+ GitHub stars**.

AYRA — blog.myosa-sensors.org/blog.html?file=blogs/ayra.md
TinyML-Based Autonomic Dysreflexia Early Warning System
- Built a below-knee early-warning wearable for autonomic dysreflexia, combining **GSR, PPG/HRV, ECG, skin temperature, humidity, and motion** with an **int8-quantized TinyML model** on ESP32.
- Presented AYRA at the **IEEE BioSensors 2026** conference in Bangkok, Thailand.

NeuroLearn — github.com/arxhr007/neurolearn
Adaptive AI Tutor
- Built an adaptive tutor for neurodivergent learners using LangGraph, personalized learning flows, and RAG.
- Presented at an OpenAI Community Meetup.

Amma — github.com/arxhr007/amma
Malayalam Programming Language
- Designed a Rust interpreter (lexer, parser, evaluator) with transliteration for Malayalam script or romanized English.

Open Source Contributions

- **ggml-org/llama.cpp (Python):** Rejected malformed `gguf-py` tokenizer merge pairs to prevent corrupted models.
- **apache/kafka (Java) & apache/airflow (Python):** Rejected negative wire-decoded field sizes reaching `new byte[size]`; fixed a secrets masker leaking a prefixed secret's tail into logs.
- **docker/docker-agent (Go):** Fixed safety rules that let `sort -o` and `tree -o` overwrite files without a prompt; added regression tests.
- **redis/redis-py (Python):** Fixed `CLIENT LIST` parsing splitting records on VT, FF, NEL and U+2028; unified both parse paths.
- **projectdiscovery nuclei, httpx & subfinder (Go):** Fixed template validation, charset decoding, and `-max-results` limits.
- **awslabs/llrt & VirusTotal/yara-x (Rust):** Aligned `Buffer` encodings and `concat` with Node.js; let `-ignore-module` skip partially implemented modules.
- **radareorg/radare2 & fastfetch (C):** Added a histogram ruler, fixed `Cs` arguments, and three format-parser bugs.
- **Also contributed:** python-pillow/Pillow, bettercap/bettercap, TabularisDB/tabularis, mvt-project/mvt, awslabs/amazon-documentdb-tools.

Leadership

- **Coders Club Lead, IEDC Sahrdaya:** Organized developer workshops, project sessions, and innovation activities.
- **Technical Team, IEEE Sahrdaya:** Built event websites, attendance systems, and event management applications.

Education

B.Tech in Computer Science and Engineering — **Kodakara, KL**
APJ Abdul Kalam Technological University (KTU) — *2023 – 2027*

Achievements

- **AYRA:** MYOSA International Hackathon finalist; presented at IEEE BioSensors 2026, Bangkok, Thailand.
- **4x CTF Winner:** Won four Capture The Flag cybersecurity competitions.